

The first three source costs from certified theta moments

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This proof is an addendum to *Certified higher theta moments and the original gamma arithmetic coefficients*. References GS.1–GS.40 refer to the full preceding proof, delivered alongside this addendum.

1 The first three original source costs at every tensor degree

Retain the exact theta seed, its Fourier coordinate and its measure

$$w(t) dt = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt.$$

Equations (GS.1)–(GS.18) prove that w is even, has positive mass μ_0 , and has finite moments of every order. The certificate of (GS.36)–(GS.37) encloses the original $\mu_0, \mu_2, \mu_4, \mu_6$. This section transports those existing certified integrals by exact rational arithmetic. It performs no further quadrature.

For each integer $k \geq 1$ put $u = t_1 + \cdots + t_k$, $m_k = w^{*k}$ and $M_j(k) = \int u^j m_k(u) du$. Fubini and evenness give the full first four even sum moments

$$\begin{aligned} M_0(k) &= \mu_0^k, \\ M_2(k) &= k\mu_2\mu_0^{k-1}, \\ M_4(k) &= k\mu_4\mu_0^{k-1} + 3k(k-1)\mu_2^2\mu_0^{k-2}, \\ M_6(k) &= k\mu_6\mu_0^{k-1} + 15k(k-1)\mu_4\mu_2\mu_0^{k-2} + 15k(k-1)(k-2)\mu_2^3\mu_0^{k-3}. \end{aligned} \tag{GS.41}$$

Here every missing term integrates to zero because at least one factor has an odd moment. For M_4 , the surviving exponent partitions are 4 and $2 + 2$. The second has $\binom{k}{2} 4!/(2!2!) = 3k(k-1)$ terms. For M_6 the partitions are 6, $4 + 2$ and $2 + 2 + 2$; their numbers of terms are respectively k , $k(k-1)6!/(4!2!) = 15k(k-1)$ and $\binom{k}{3} 6!/(2!2!2!) = 15k(k-1)(k-2)$. The second-moment expansion has precisely k surviving terms. These finite counts prove (GS.41) with its full mass factors. Because $\mu_0 > 0$, the displayed powers are defined even when a zero combinatorial coefficient accompanies a negative exponent.

All nonzero polynomial norms in $L^2(m_k du)$ are strictly positive. Indeed the nonnegative density m_k has positive mass μ_0^k , so its positivity set has positive Lebesgue measure. A nonzero polynomial has finitely many real roots, as repeated division by its distinct linear factors cannot exceed its degree. Its squared modulus times m_k is therefore positive on a set of positive measure, and its integral is positive. This proves every positivity and division below for the actual source.

Put $\nu = \mu_2/\mu_0$. With the norms taken in the unchanged measure $m_k du$, the first four monic orthogonal polynomials and all their squared norms are

$$\begin{aligned} p_0(u) &= 1, & h_0(k) &= M_0(k), \\ p_1(u) &= u, & h_1(k) &= M_2(k), \\ p_2(u) &= u^2 - k\nu, & h_2(k) &= M_4(k) - \frac{M_2(k)^2}{M_0(k)}, \\ p_3(u) &= u^3 - \frac{M_4(k)}{M_2(k)}u, & h_3(k) &= M_6(k) - \frac{M_4(k)^2}{M_2(k)}. \end{aligned} \quad (\text{GS.42})$$

To verify orthogonality of p_2 , its pairing with 1 is $M_2 - k\nu M_0 = 0$, and its pairing with u is zero by evenness. For p_3 , parity gives zero against 1 and p_2 ; its pairing with u is $M_4 - (M_4/M_2)M_2 = 0$. Expanding each squared polynomial gives the norms in (GS.42). Uniqueness follows because the difference of two monic orthogonal polynomials of the same degree lies in the lower-degree space, is orthogonal to itself, and hence is zero by the preceding positivity argument.

Define the positive original monic source costs $a_j^{(k)} = h_j(k)/h_{j-1}(k)$. These are exactly the lower coefficients in the real-coordinate three-term recurrence: in up_j , its pairing with p_{j-1} equals the pairing of p_j with up_{j-1} , which is h_j because $up_{j-1} - p_j$ has lower degree. Pairings with lower degrees vanish by the same argument. The degree- j coefficient vanishes by evenness. Therefore $up_j = p_{j+1} + a_j^{(k)}p_{j-1}$. For $j = 3$, the next monic polynomial p_4 exists by finite Gram orthogonalization, whose Gram is positive definite by the proved polynomial norm property; its full formula is not needed to determine the already specified coefficient h_3/h_2 . Substituting (GS.41) into (GS.42) gives

$$\begin{aligned} a_1^{(k)} &= k\nu, \\ a_2^{(k)} &= \frac{\mu_4}{\mu_2} + (2k-3)\nu, \\ a_3^{(k)} &= \frac{M_6(k) - M_4(k)^2/M_2(k)}{M_4(k) - M_2(k)^2/M_0(k)}. \end{aligned} \quad (\text{GS.43})$$

For the second equality one obtains $M_4/M_2 = \mu_4/\mu_2 + 3(k-1)\nu$ and $M_2/M_0 = k\nu$, whose exact difference is the displayed cost. For an explicit expanded version of the third equality, write $r_4 = \mu_4/\mu_0$ and $r_6 = \mu_6/\mu_0$, solely as the indicated scalar quotients of the original moments. Then

$$a_3^{(k)} = \frac{r_6 - r_4^2/\nu + 9(k-1)r_4\nu + 3(k-1)(2k-7)\nu^3}{r_4 + (2k-3)\nu^2}. \quad (\text{GS.44})$$

To check every term, divide the numerator of (GS.43) by $k\mu_0^k$ using (GS.41). The square term contributes $r_4^2/\nu + 6(k-1)r_4\nu + 9(k-1)^2\nu^3$; subtracting it from $r_6 + 15(k-1)r_4\nu + 15(k-1)(k-2)\nu^3$ gives the numerator in (GS.44). The denominator of (GS.43), divided by the same $k\mu_0^k$, is $r_4 + (2k-3)\nu^2$. Thus both original norms and all mass factors are recovered from these scalar abbreviations.

The original spectral coordinate is $S = k/2 + iu$. Let $c_k = k/2$ and define the algebra isomorphism

$$\Psi_k : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad (\Psi_k P)(u) = P(c_k + iu), \quad \Psi_k^{-1}p(S) = p((S - c_k)/i). \quad (\text{GS.45})$$

Direct substitution proves the two inverse identities. Give $\mathbb{C}[S]$ the source norm $\int |P(c_k + iu)|^2 m_k(u) du$; then Ψ_k is a linear isometry onto the polynomial source in the real coordinate. On the degree- j monic affine space, its phase-preserving monic inverse is $p \mapsto P_j = i^j \Psi_k^{-1}p$. Since

$|i^j| = 1$, this map preserves the original norm and has inverse $P \mapsto i^{-j} \Psi_k P$ on that affine space. In particular the complete first four spectral monic polynomials are

$$\begin{aligned} P_0 &= 1, & P_1 &= S - c_k, & P_2 &= (S - c_k)^2 + k\nu, \\ P_3 &= (S - c_k)^3 + \left\{ \frac{\mu_4}{\mu_2} + 3(k-1)\nu \right\} (S - c_k). \end{aligned} \quad (\text{GS.46})$$

Their squared norms are exactly h_0, h_1, h_2, h_3 from (GS.42). Multiplying the real recurrence by i^{j+1} proves the spectral recurrence with its actual lower sign,

$$SP_j = P_{j+1} + c_k P_j - a_j^{(k)} P_{j-1}. \quad (\text{GS.47})$$

Thus the positive norm ratio $a_j^{(k)}$ and the negative lower coefficient in the S presentation have their exact phase map.

To connect these costs to the existing Gamma determinant corrections, let D_n be the determinant of the source Gram on $1, u, \dots, u^{n-1}$, with $D_0 = 1$, and let D_n^Γ use the Gamma sum measure $r_{13/4}^{*k} du$. Its mass is $c_{13/4}^k$, and its monic norms are $c_{13/4}^k j! (13k/2)_j$, by (GS.23), Fourier uniqueness and (GS.28). Specifically Fourier transformation of the convolution gives $c_{13/4}^k \cosh^{-13k/2} y$; division by the mass constant in (GS.23) identifies its density as a positive scalar times $r_{13k/4}$, proving the asserted norms without any change to u . Monic orthogonalization is triangular with diagonal one, so $D_n = \prod_{j < n} h_j$ and the analogous Gamma product holds. Define $X_n = D_n / D_n^\Gamma$ and

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad a_{N+1}^{(k)} = (N+1)(N+13k/2) Q_N. \quad (\text{GS.48})$$

Dividing the two consecutive determinant products proves the second identity. In particular, with $\eta = \mu_4 / \mu_2$,

$$\begin{aligned} Q_0 &= \frac{2\nu}{13}, \\ Q_1(k) &= \frac{\eta + (2k-3)\nu}{13k+2} = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \eta - \frac{43}{13}\nu, \\ Q_2(k) &= \frac{2a_3^{(k)}}{39k+12}. \end{aligned} \quad (\text{GS.49})$$

Every equality is obtained by substituting (GS.43) into (GS.48). No volume of a nonzero packet quotient enters these source costs.

The pure rational checker reads the exact dyadic moment endpoints from the pinned GS.36 certificate and propagates them through (GS.43)–(GS.49) using `fractions.Fraction` exclusively. For intervals $[a, b]$ and $[c, d]$, it uses endpoint addition, $[a - d, b - c]$ for subtraction, the minimum and maximum of the four endpoint products for multiplication, and multiplication by $[1/d, 1/c]$ for division after checking $c > 0$. These rules follow from linearity in each variable and the decreasing map $x \mapsto 1/x$ on the positive axis. Thus they preserve inclusion without any floating-point

operation. Applying (GS.37) with exact rational floor and ceiling yields, among the saved intervals,

$$\begin{aligned}
\nu &\in [10.2075647534857216888657612522375551749964, \\
&\quad 10.2075647534857216888657612522375551749967], \\
Q_0 &\in [1.5703945774593417982870401926519315653839, \\
&\quad 1.5703945774593417982870401926519315653842], \\
a_2^{(1)} &\in [19.1496329281085602472067784658376980738791, \\
&\quad 19.1496329281085602472067784658376980738794], \\
\rho &\in [-4.4062857337815667270988244239412754068822, \\
&\quad -4.4062857337815667270988244239412754068819].
\end{aligned}$$

All four widths are $3 \cdot 10^{-40}$. The receipt also encloses μ_4/μ_2 , $\mu_4\mu_0 - \mu_2^2$, $a_2^{(2)}$, $Q_1(1)$, $a_3^{(1)}$ and $Q_2(1)$. In particular $\rho < 0$ and (GS.49) give the exact fixed-degree all- k comparison

$$0 < Q_1(1) \leq Q_1(k) < Q_0, \quad Q_1(k+1) - Q_1(k) = \frac{-13\rho}{(13k+2)(13k+15)} > 0. \quad (\text{GS.50})$$

The positive lower bound follows either from $h_2/h_1 > 0$ or its certified $k = 1$ enclosure; the difference follows by literal subtraction in (GS.49). The computation concerns original source polynomial degrees at most three, at every specified tensor degree. The analytic seed still has quotient $\mathbb{C}[S]/(1) = 0$, its quotient determinant is the empty determinant one, and every nonzero source norm above remains before that zero quotient.